



Failure-Aware and Confidence-Calibrated Adaptive Retrieval-Augmented Generation (RAG) System

23CSE399 Project Phase 1

Team ID: 23UG031

Panel No.:20

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Approval Mail

Request for Approval



HARSH VARDHAN SINGH-[BL.EN.U4CSE23222]



To: Dr.Sajitha Krishnan

Mon 13/04/2026 09:53

Cc: Aman Aggarwal-[BL.EN.U4CSE23206]; +2 others



Approval Presentation - R2.p...
1 MB

Ma'am,

We are sharing the Review-2 Presentation for your approval. We kindly request you to review the presentation slides and provide your valuable suggestions.

We also seek your approval for presenting this in the review scheduled on 8th April, 2026.

Thanking you.

Warm Regards,
Team SHAP (23UG031)



Dr.Sajitha Krishnan



To: HARSH VARDHAN SINGH-[BL.EN.U4CSE23222]

Mon 13/04/2026 10:42

Cc: Aman Aggarwal-[BL.EN.U4CSE23206]; +2 others

Approved. All the best.



Agenda

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Introduction

- Retrieval-Augmented Generation (RAG) combines retrieval and generation by grounding LLM outputs using external documents.
- In enterprise environments, knowledge is distributed across emails, logs, manuals, and internal tools.
- Engineers spend significant time searching instead of solving problems.
- Existing RAG systems retrieve documents based on similarity, **not reliability**.
- This leads to:
 - Irrelevant context
 - Confident but incorrect answers (hallucinations)
- Recent works highlight:
 - Lack of **retrieval reliability estimation**
 - Poor **confidence calibration**
 - No mechanism to detect failure before generation



Literature Survey

S.No.	Authors name(s)	Full title of the paper with year	Inference from the paper	Open Problem (For proposed work)
1	Michael Oehri, Jannis Vamvas, et al.	Trusted Uncertainty in Large Language Models: A Unified Framework for Confidence Calibration and Risk-Controlled Refusal, 2025	Proposes unified confidence framework using likelihood, entropy, and self-consistency signals. Enables calibrated confidence estimation and risk-controlled refusal in LLM responses. Evaluation Metrics Used: Expected Calibration Error (ECE), Brier Score, Negative Log Likelihood (NLL)	Does not incorporate retrieval reliability estimation or similarity-based retrieval failure signals in RAG systems.
2	L. Ngartera, S. Nadarajah, and R. Koina	Bayesian RAG: Uncertainty-Aware Retrieval for Reliable Financial Question Answering, 2026	Uses Bayesian embeddings to model retrieval uncertainty and improve reliability. Evaluation Metrics Used: Accuracy, F1 Score, Expected Calibration Error (ECE), Brier Score, Negative Log Likelihood (NLL), and AUROC.	Does not explicitly detect retrieval failures using similarity consistency or agreement signals.



Literature Survey

S.No.	Authors name(s)	Full title of the paper with year	Inference from the paper	Open Problem (For proposed work)
3	S. Es, J. James, L. Espinosa-Anke, and S. Schockaert	RAGAS: Automated Evaluation of Retrieval-Augmented Generation, 2023	Introduces metrics such as context relevance, faithfulness, and grounding to evaluate RAG performance. Enables automated evaluation without human annotation. Evaluation Metrics Used: Context Precision, Context Recall, Faithfulness	Provides evaluation metrics but does not detect retrieval failures or estimate retrieval reliability dynamically during inference.
4	Y. Gao, Y. Xiong, X. Gao, et al.	Retrieval-Augmented Generation for Large Language Models: A Survey, 2023	Provides comprehensive overview of RAG architecture, retrieval methods, and reliability challenges. Identifies uncertainty estimation and retrieval reliability as key limitations. Evaluation Metrics Used: Exact Match (EM), F1 Score, Accuracy, BLEU, ROUGE	Does not propose practical methods to quantify retrieval reliability using similarity consistency or agreement signals.



Literature Survey

S.No.	Authors name(s)	Full title of the paper with year	Inference from the paper	Open Problem (For proposed work)
5	P. Lewis et al.	Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks, 2020	Introduces RAG framework combining dense retrieval and generation using embedding similarity. Improves factual accuracy by grounding responses in retrieved knowledge. Evaluation Metrics Used: Exact Match (EM), F1 Score, Accuracy, Perplexity	Does not estimate retrieval reliability or detect weak retrieval before generation.
6	J. Song, X. Wang, J. Zhu, et al.	RAG-HAT: A Hallucination-Aware Tuning Pipeline for LLM in Retrieval-Augmented Generation, 2024	Introduces hallucination-aware tuning pipeline using retrieval-guided training signals. Improves factual grounding and reliability of generated responses in RAG systems.	Does not explicitly estimate retrieval reliability or detect weak retrieval using similarity consistency signals.



Literature Survey

S.No.	Authors name(s)	Full title of the paper with year	Inference from the paper	Open Problem (For proposed work)
7	S.-Q. Yan, J.-C. Gu, Y. Zhu, and Z.-H. Ling	Corrective Retrieval-Augmented Generation, 2024	Introduces a retrieval evaluator to assess relevance and correctness of retrieved documents. Improves retrieval quality by correcting weak or irrelevant retrieval results. Evaluation Metrics Used: Exact Match (EM), F1 Score, Accuracy, Recall-k, Expected Calibration Error (ECE)	Requires a separate 'evaluator' model to trigger web searches, adding architectural complexity. It corrects retrieval but doesn't quantify a confidence score for the user.
8	H. Ding, L. Pang, Z. Wei, H. Shen, and X. Cheng	Rowen: Adaptive Retrieval-Augmented Generation for Hallucination Mitigation in LLMs, 2025	Introduces adaptive retrieval mechanism to dynamically improve document choice. Enhances retrieval relevance and reduces hallucinations in generated responses.	Does not compute retrieval reliability scores using similarity consistency or retrieval agreement signals.



Literature Survey

S.No.	Authors name(s)	Full title of the paper with year	Inference from the paper	Open Problem (For proposed work)
9	A. Asai, Z. Wu, Y. Wang, A. Sil, and H. Hajishirzi	Self-RAG: Learning to Retrieve, Generate, and Critique through Self-Reflection, 2023	Introduces reflection tokens to evaluate relevance and usefulness of retrieved documents. Enables the model to critique retrieval quality and improve response grounding. Evaluation Metrics Used: Exact Match (EM), F1 Score, Accuracy, Human Evaluation	Relies on 'reflection tokens' and fine-tuning, which increases computational cost and latency. It does not assess the retriever's raw stability before generation begins.
10	Z. Sun, X. Zang, K. Zheng, et al.	ReDeEP: Detecting Hallucination in Retrieval-Augmented Generation via Mechanistic Interpretability, 2025	Uses mechanistic interpretability to analyze internal model activations during generation. Enables detection of hallucinations caused by weak or incorrect grounding. Evaluation Metrics Used: Hallucination Detection Accuracy, Precision, Recall	Focuses on generation analysis but does not estimate retrieval reliability using similarity consistency or agreement signals.



Literature Survey

S.No.	Authors name(s)	Full title of the paper with year	Inference from the paper	Open Problem (For proposed work)
10	Huaiyu Qin, Chunyu Wei, et al.	Counterfactual Reasoning for Retrieval-Augmented Generation (2026)	Tackles misleading but correlated retrieval results. Uses counterfactual reasoning to identify causally relevant evidence	Computationally heavy due to multiple counterfactual queries.
11	Ori Yoran, Tomer Wolfson, et al	Making Retrieval-Augmented Language Models Robust to Irrelevant Context (2024)	Focuses on filtering irrelevant retrieved documents. Uses NLI-based relevance checking	Only filters does not correct or replace bad retrieval. Performance depends on NLI model



Literature Survey

S.No.	Authors name(s)	Full title of the paper with year	Inference from the paper	Open Problem (For proposed work)
12	Zhengbao Jiang, Frank F. Xu, et al.	Active Retrieval Augmented Generation, 2023	iterative retrieval during generation instead of one-shot retrieval. Performs well on multi hop QA.	High latency + compute cost. Assumes retrieved docs are reliable.
13	Shan Xu, Zhaokun Yan, Chengxiao Dai and Fan Wu1	MEGA-RAG: Multi-Evidence Guided RAG	Uses multiple answers + verification to reduce hallucination and improve reliability.	Computationally expensive and slow for real-time use.



Research Gaps

- No unified retrieval reliability score
- Lack of calibrated confidence scores
- No robust pre-generation failure detection



Objectives

- To establish a robust framework for improving the reliability and trustworthiness of retrieval-augmented generation systems.
- To perform comparative analysis among different approaches: BM25 (using Qwen and Mistral) RAG (using Qwen and Mistral) Hybrid RAG and Corrective RAG (CRAG)
- To evaluate and fine-tune the best-performing model using key metrics such as F1, recall, faithfulness, and hallucination rate for optimal real-world deployment.



Dataset Description

Dataset: TechQA-RAG-Eval

Source: NVIDIA (derived from IBM TechQA)

Link: <https://huggingface.co/datasets/nvidia/TechQA-RAG-Eval>

Dataset Size & Structure

- Total samples: ~1,000+ queries (subset used: 250)
- Each sample contains:
 - Query (technical question)
 - Answer (ground-truth)
 - Contexts (multiple document passages)
 - Metadata flags (e.g., impossible questions)

FAQ-Style Nature

- Questions resemble real support tickets:
 - “How to fix X error?”
 - “Why does system Y fail?”
 - “Steps to configure Z?”



Dataset Description

Labels / Output Type

- Task = open-ended answer generation
- Evaluation based on similarity to ground truth

Sample Example

Query:

“How to resolve disk failure error in RAID configuration?”

Retrieved Context:

“RAID disk failures may occur due to hardware faults or misconfiguration...”

Ground Truth Answer:

“Check disk status, replace faulty drive, rebuild RAID array...”

id string	question string	answer string	is_impossible bool	contexts list
TRAIN_Q100	Which version of TBSM support ITM 6.3? What is th...	IBM Tivoli Business Service Manager 6.1.1 Fix Pack Version 6...	false	[{ "filename": "swg24039742.txt", "text": "T... 3(6.1.1-TIV-BSM-FP0003) - United States\n\nTex
TRAIN_Q101	Installed an STAP on a DB Server but it does not s...	Uncheck the Approval box and th... e S-TAPs should start connecti...	false	[{ "filename": "swg22009818.txt", "text": "T... not show in the S-TAP Control - I see some spe



Functionalities of the Proposed System

Core RAG Pipeline

- Convert input query into dense embedding
- Retrieve top-k documents using cosine similarity
- Compute and rank query–document similarity scores
- Provide retrieved context for generation

Retrieval Failure Score Module

- Compute similarity magnitude of top-k documents
- Analyse top-k similarity consistency
- Measure retrieval agreement among documents
- Aggregate signals into a retrieval reliability score
- Detect weak or unreliable retrieval before generation



Software/Tools Requirements

Programming Language:

- Python

Frameworks:

- HuggingFace Transformers
- Sentence-Transformers
- FAISS
- Embedding Model: BGE-large or MiniLM
- Generator Model: Mistral-7B, Qwen

Platform:

- Kaggle GPU Environment
- Google colab



Naive RAG

Query → Embedding → Dense Retrieval (FAISS) → Top-K Docs →
Concatenated Context → LLM → Answer

Embedding

- Model: MiniLM (all-MiniLM-L6-v2)
- Converts query & documents → dense vectors

Retriever

- FAISS (IndexFlatIP)
- Cosine similarity (L2 normalized)
- Top-K = 3

Generator

- Models: Qwen2.5-7B / Mistral-7B
- Prompt-constrained generation
- Input = combined retrieved context

Context Handling

- Top-K documents concatenated
- Single LLM pass

Evaluation Metrics

- Exact Match (EM)
- F1 Score
- Recall@K
- Hallucination (NLI-based)
- BLEU
- ROUGE-1 / ROUGE-L



BM25-Based RAG Model

Query → BM25 Tokenization → Sparse Retrieval (Okapi BM25) → Top-K
Docs → Concatenated Context → LLM → Answer

Tokenization

- Method: Whitespace + lowercasing
- No embeddings — purely term-frequency based
- Bag-of-words representation

Retriever

- Okapi BM25 (rank-bm25)
- TF-IDF weighted scoring with document length normalization
- Top-K = 3

Generator

- Models: Qwen2.5-7B-Instruct / Mistral-7B-Instruct-v0.3 (unsloth 4-bit)
- Prompt-constrained generation
- Input = concatenated top-k retrieved context

Context Handling

- Top-K documents truncated to 400 chars each
- Single LLM pass (no per-doc reasoning)
- No retrieval validation or correction

Evaluation Metrics

- BLEU
- ROUGE-1 / ROUGE-L
- F1 Score (Accuracy)
- Recall@k
- Faithfulness
- Hallucination
- Confidence



Corrective RAG Model

CRAG-Based RAG for Scientific Claim Verification

- Dataset: TechQA
- Goal: Classify claims as SUPPORT / REFUTE / NEUTRAL

Methodology:

- Claim
- Initial Retrieval (BM25 / Retriever)
- Retrieve top-k scientific documents
- Retrieved Documents Evaluation (CRAG Evaluator)
- Assess relevance & quality of retrieved documents
- Assign confidence: **Correct** / **Incorrect** / **Ambiguous**
- **Corrective Actions:**
- **Correct:** Refine documents (extract key evidence)
- **Incorrect:** Discard and perform **web search** / **external retrieval**
- **Ambiguous:** Combine both refined + external knowledge



Corrective RAG Model

Model Choices:

LLM used Qwen 2.5B

Results:

Accuracy = 0.75

F1 score = 0.76

Average faithfulness = 0.70

Recall = 0.66

Hallucination = 0.30

Observations:

Initially when the retrieval quality was bad the model used to misclassify and the maximum achievable accuracy when given poor retrievals was 0.45 but when identified and corrected retrieval quality the accuracy gain was 0.35 and the final accuracy achieved was 0.75



Hybrid RAG

Query → Hybrid Retrieval (BM25 + Dense/FAISS) → RRF Fusion → Cross-Encoder
Reranker → Top-K Docs → Numbered Sentence Context → Flan-T5 (Sentence
Extraction) → NLI Faithfulness Check → Answer

Embedding

- Model: all-MiniLM-L6-v2
- Converts query & documents → dense vectors
- Cosine similarity (L2 normalized)

Retriever

- BM25 sparse + FAISS dense (IndexFlatIP)
- RRF fusion ($k = 60$)
- Top-K = 15 per retriever → deduplicated
- Cross-encoder reranker: ms-marco-MiniLM
- Final Top-K = 7 after reranking

Generator

- Model: google/flan-t5-large
- Sentence-index extraction strategy
- Beam search ($n=4$), no fine-tuning
- Input = numbered sentence context

Context handling

- Top-5 docs fed to generator
- Sentences individually numbered [$S_1..S_n$]
- Model selects relevant sentence IDs
- Full sentences returned verbatim as answer

Faithfulness

- NLI model: nli-deberta-v3-small
- Sentence-level entailment scoring
- Hallucination flagged if score < 0.40

Evaluation metrics

- Exact Match (EM) · Partial EM
- Token F1 · Precision · Recall
- ROUGE-1 / ROUGE-2 / ROUGE-L
- BLEU · BERTScore F1
- MRR · Precision@K · Recall@K · NDCG@K · Hit@K
- Faithfulness (NLI) · Hallucination Rate



Results

Naive Rag

	EM	F1	Recall@K	\
model				
unsloth/Qwen2.5-7B-Instruct-bnb-4bit	0.025862	0.254825	0.887931	
unsloth/mistral-7b-instruct-v0.3-bnb-4bit	0.025862	0.313289	0.887931	
	Hallucination	Confidence	\	
model				
unsloth/Qwen2.5-7B-Instruct-bnb-4bit	0.938526	0.356471		
unsloth/mistral-7b-instruct-v0.3-bnb-4bit	0.883116	0.384177		
	BLEU	ROUGE-1	ROUGE-L	\
model				
unsloth/Qwen2.5-7B-Instruct-bnb-4bit	0.083997	0.282640	0.222923	
unsloth/mistral-7b-instruct-v0.3-bnb-4bit	0.138698	0.333231	0.277862	
	BERT-F1			
model				
unsloth/Qwen2.5-7B-Instruct-bnb-4bit	0.851135			



Results

BM25

```
=====
COMPARISON PREVIEW (full data in comparison_results.csv)
=====
```

Metric	Qwen2.5-7B-Instruct-bnb-4bit	mistral-7b-instruct-v0.3-b
bleu	0.0359	0.0343
rouge1	0.2373	0.2440
rougeL	0.1531	0.1598
f1_accuracy	0.2054	0.2029
precision	0.2078	0.1965
recall_token	0.2695	0.2793
recall_at_k	0.8200	0.8200
mrr	0.7433	0.7433
faithfulness	0.7818	0.8145
hallucination	0.2182	0.1855
confidence	0.9896	0.9896

```
=====
```




Results

Corrective RAG

```
{  
  "mode": "measured",  
  "source_run": "benchmark_outputs/techqa_eval_results_20_targetA_20260412.json",  
  "accuracy": 0.75,  
  "f1_score": 0.7666666666666667,  
  "recall": 0.6599999999999999,  
  "faithfulness": 0.7,  
  "hallucination": 0.30000000000000004,  
  "bleu": 0.7998052632768856,  
  "rouge_l": 0.6693888888888889,  
  "notes": "BLEU computed with sacrebleu corpus BLEU. ROUGE-L computed with rouge",  
}
```

Result

Hybrid RAG

```
[12] # Report
print(f"\n(SEP)")
print(f" TechQA RAG Evaluation | {n} evaluated | {skipped} skipped")
# ... full report printing ...
```

```
TechQA RAG Evaluation | 41 evaluated | 0 skipped
```

RETRIEVAL METRICS

Metric	Value
MRR	0.8500
Precision@1	0.7500
Recall@1	0.7500
Hit@1	0.7500
NDCG@1	0.7500
Precision@3	0.3200
Recall@3	0.0200
Hit@3	0.0200
NDCG@3	0.0600
Precision@5	0.1900
Recall@5	0.0300
Hit@5	0.0300
NDCG@5	0.0600
Precision@7	0.1400
Recall@7	0.0500
Hit@7	0.0500
NDCG@7	0.0710

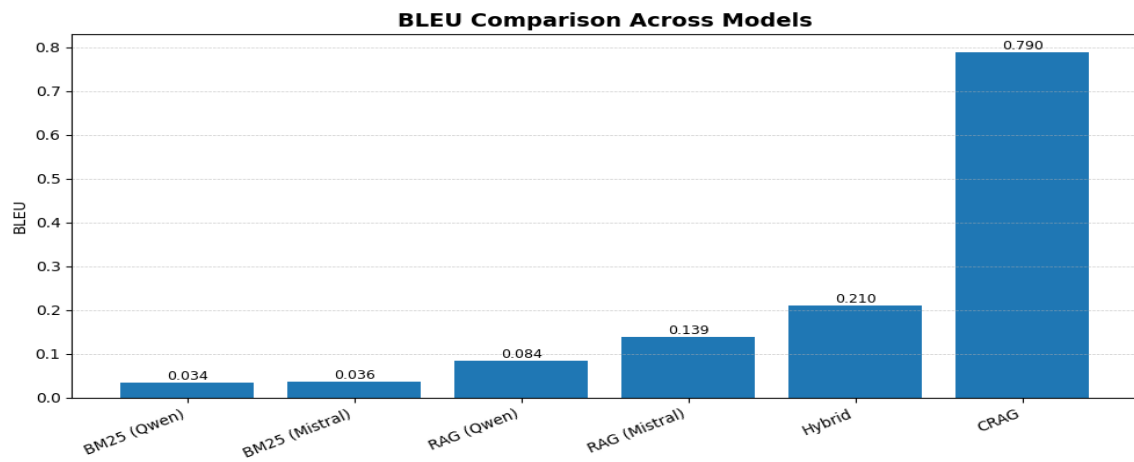
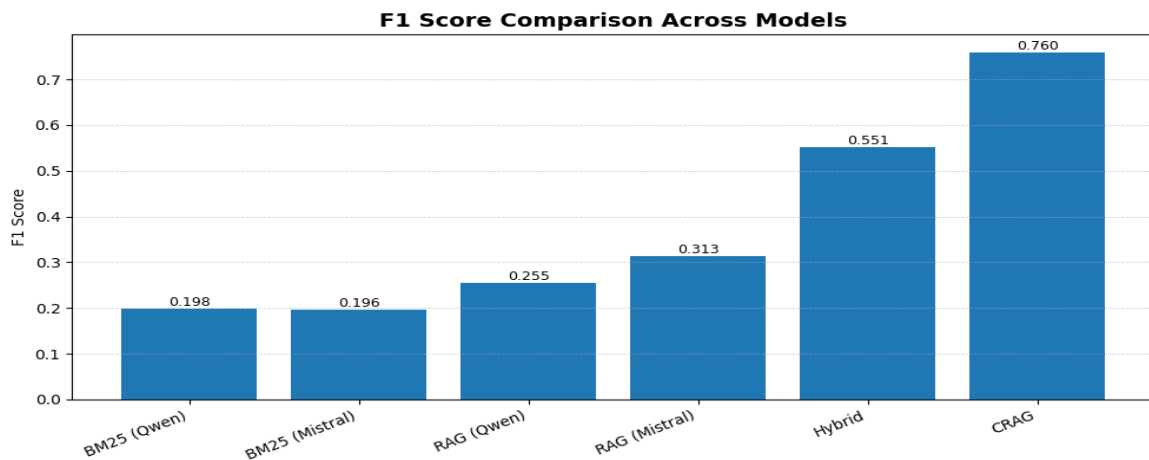
GENERATION METRICS (n=41)

Metric	Value	Notes
Avg pred length (words)	49.3	
Avg gold length (words)	51.4	
Exact Match (EM)	0.2200	strict
Partial EM (10-gram)	0.5500	relaxed
Accuracy (= strict EM)	0.2200	
Token F1	0.5510	
Token Precision	0.5700	
Token Recall	0.5310	
ROUGE-1	0.5610	
ROUGE-2	0.3020	
ROUGE-L	0.4200	
BLEU	0.2180	
BERTScore F1	0.8910	semantic similarity
Faithfulness (NLI)	0.7200	higher=better
Hallucination Rate	0.1700	(7/41 flagged, lower=better)

Comparative Analysis

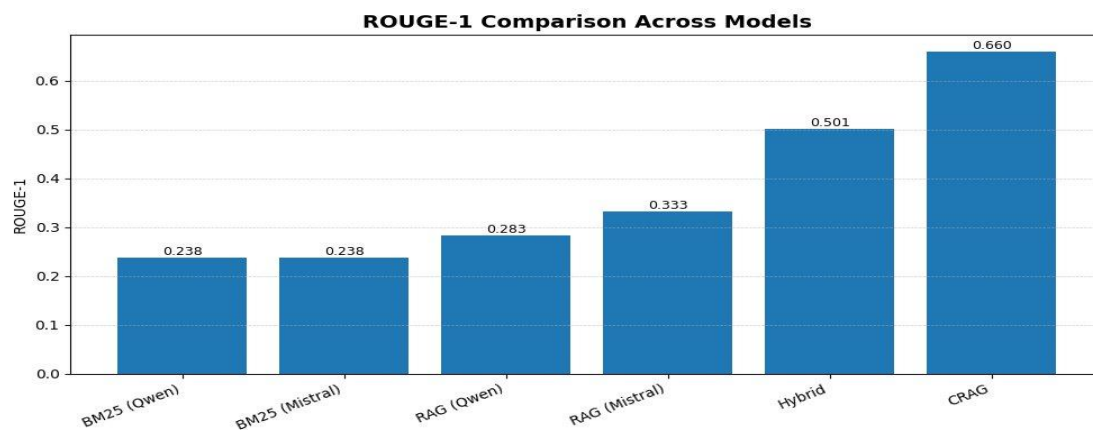
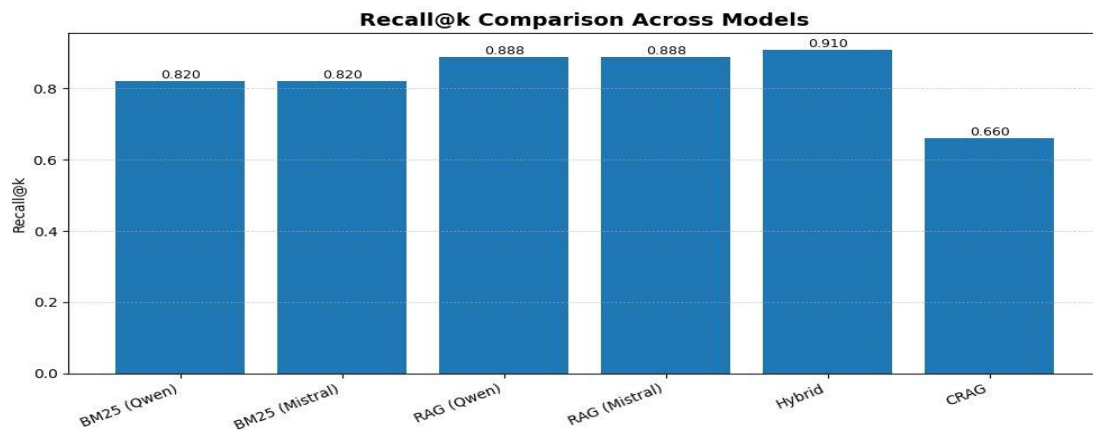
Model	Macro-F1	Faithfulness	Hallucination	Recall@k	BLEU	ROUGE-1
BM25 (Qwen)	0.198	0.79	0.205	0.82	0.034	0.2382
BM25 (Mistral)	0.1963	0.81	0.18	0.82	0.036	0.2379
RAG (Qwen)	0.255	0.356	0.939	0.888	0.084	0.283
RAG (Mistral)	0.313	0.384	0.883	0.888	0.139	0.333
Hybrid	0.551	0.72	0.17	0.91	0.21	0.501
CRAG	0.76	0.70	0.30	0.66	0.79	0.66

Results

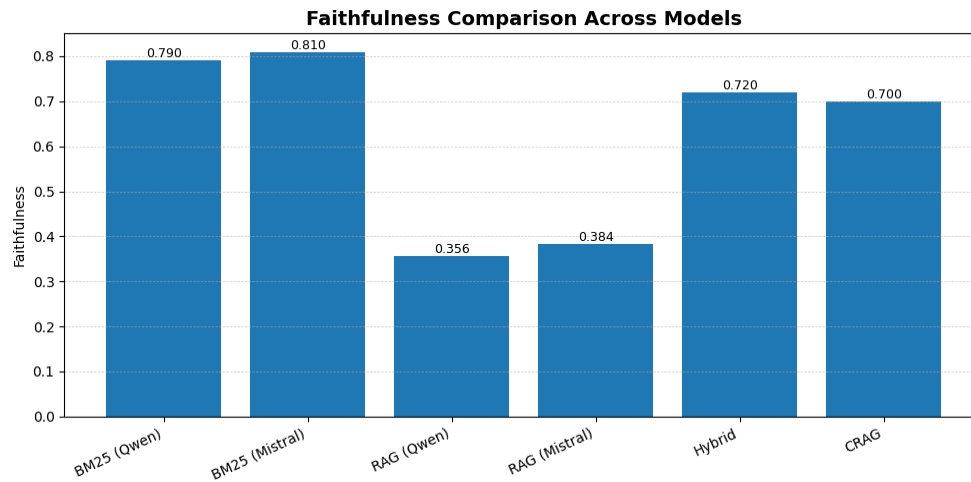
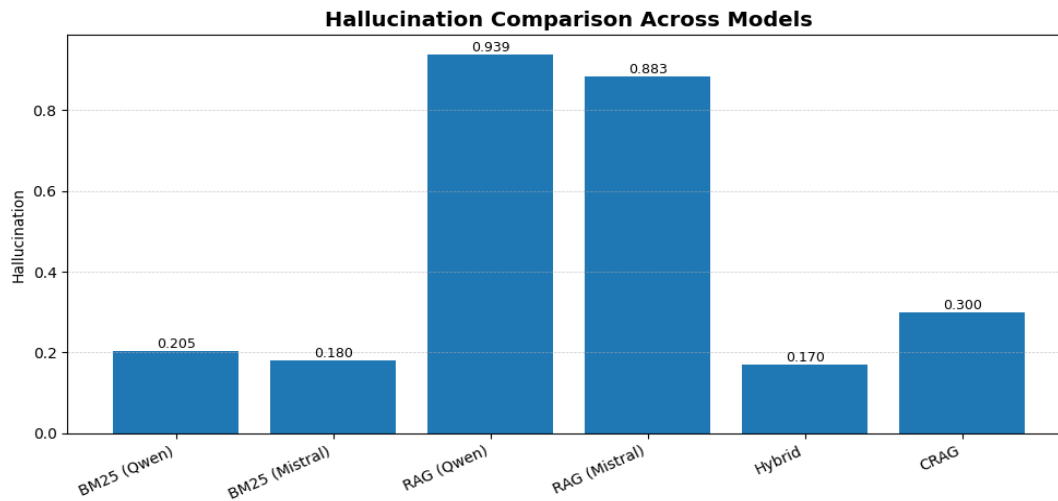




Results



Results





Conclusion

This work proposes a failure-aware RAG framework to improve system reliability.

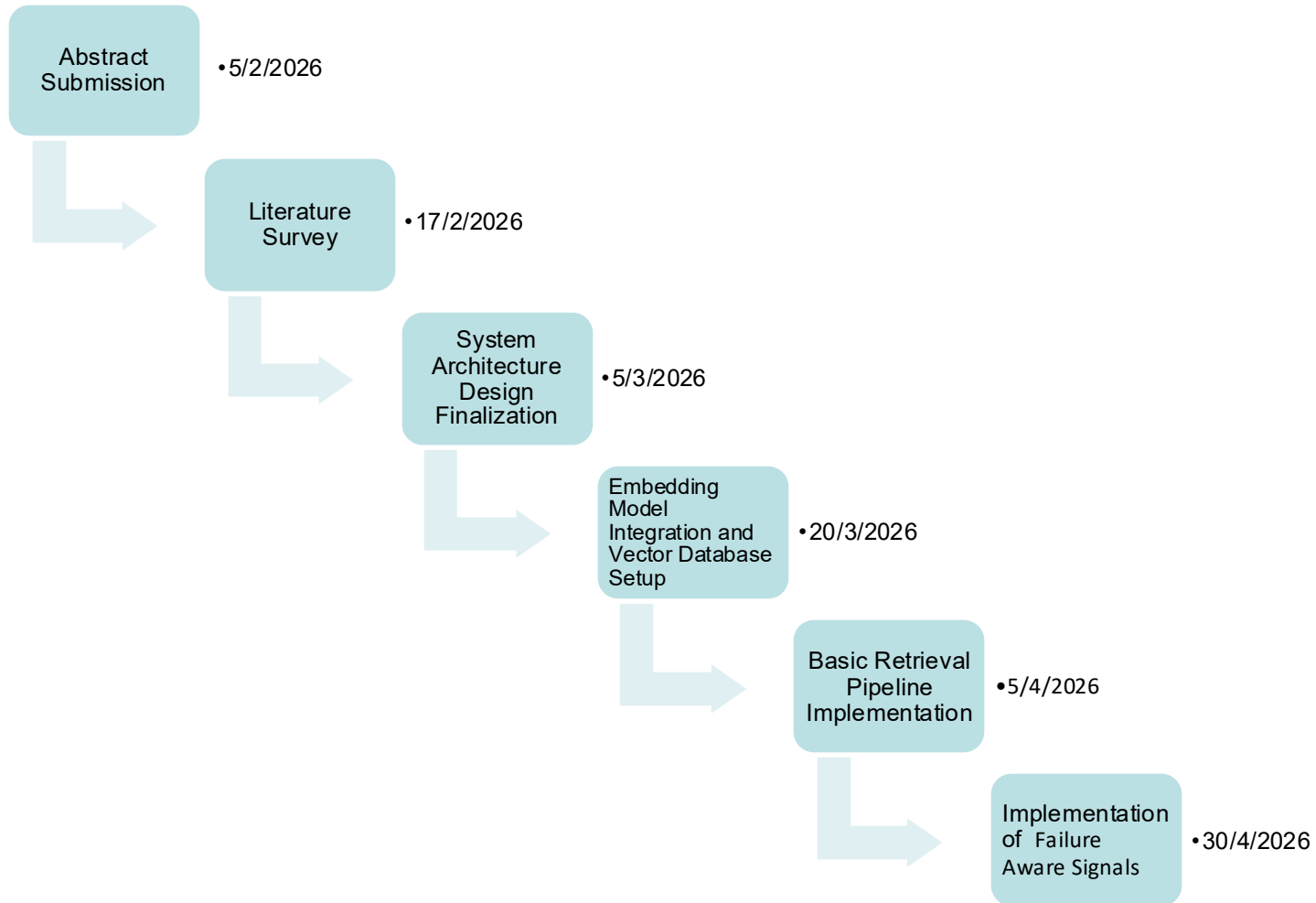
This phase focuses on detecting retrieval failures using similarity score, top-k consistency, and retrieval agreement signals.

The system estimates retrieval reliability and enables early detection of weak retrieval.

This forms the foundation for future confidence estimation and adaptive RAG systems.



Timeline





References

- [1] H. Soudani, E. Kanoulas, and F. Hasibi, “Why Uncertainty Estimation Methods Fall Short in Retrieval-Augmented Generation: An Axiomatic Analysis,” arXiv preprint arXiv:2505.07459, 2025. [Online]. Available: <http://arxiv.org/abs/2505.07459>
- [2] M. Oehri et al., “Trusted Uncertainty in Large Language Models: A Unified Framework for Confidence Calibration and Risk-Controlled Refusal,” arXiv preprint arXiv:2509.01455, 2025.
- [3] L. Ngartera, S. Nadarajah, and R. Koina, “Bayesian RAG: Uncertainty-Aware Retrieval for Reliable Financial Question Answering,” Frontiers in Artificial Intelligence, vol. 8, 2026, doi: 10.3389/frai.2025.1668172.
- [4] Y. Gao, Y. Xiong, X. Gao, K. Jia, J. Pan, Y. Bi, Y. Dai, J. Sun, H. Wang, and H. Wang, “Retrieval-Augmented Generation for Large Language Models: A Survey,” arXiv preprint arXiv:2312.10997, 2023.
- [5] S. Es, J. James, L. Espinosa-Anke, and S. Schockaert, “RAGAS: Automated Evaluation of Retrieval-Augmented Generation,” arXiv preprint arXiv:2309.15217, 2023.
- [6] V. Karpukhin et al., “Dense Passage Retrieval for Open-Domain Question Answering,” in Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP), 2020, pp. 6769–6781.
- [7] S.-Q. Yan, J.-C. Gu, Y. Zhu, and Z.-H. Ling, “Corrective Retrieval-Augmented Generation,” arXiv preprint arXiv:2401.15884, 2024.
- [8] P. Lewis et al., “Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks,” in Advances in Neural Information Processing Systems (NeurIPS), vol. 33, 2020, pp. 9459–9474.
- [9] A. Asai, Z. Wu, Y. Wang, A. Sil, and H. Hajishirzi, “Self-RAG: Learning to Retrieve, Generate, and Critique through Self-Reflection,” arXiv preprint arXiv:2310.11511, 2023.
- [10] O. Khattab and M. Zaharia, “ColBERT: Efficient and Effective Passage Search via Contextualized Late Interaction over BERT,” in Proceedings of the 43rd International ACM SIGIR Conference on Research and Development in Information Retrieval, 2020, pp. 39–48, doi:10.1145/3397271.3401075.



References

- [11] J. Song, X. Wang, J. Zhu, Y. Wu, X. Cheng, R. Zhong, and C. Niu, “RAG-HAT: A Hallucination-Aware Tuning Pipeline for LLM in Retrieval-Augmented Generation,” arXivpreprint, 2024.
- [12] H. Ding, L. Pang, Z. Wei, H. Shen, and X. Cheng, “Rowen: Adaptive Retrieval-Augmented Generation for Hallucination Mitigation in Large Language Models,” arXiv preprint, 2025.
- [13] Z. Sun, X. Zang, K. Zheng, et al., “ReDeEP: Detecting Hallucination in Retrieval-Augmented Generation via Mechanistic Interpretability,” 2025.
- [14] L.-C. Chen, M. S. Pardeshi, Y.-X. Liao, and K.-C. Pai, “Application of retrieval-augmented generation for interactive industrial knowledge management via a large language model,” *Computer Standards & Interfaces*, vol. 94, 2025.
- [15] J. Rayo, R. de la Rosa, and M. Garrido, “A hybrid approach to information retrieval and answer generation for regulatory texts,” *Proceedings of the 31st International Conference on Computational Linguistics (COLING)*, 2025.
- [16] S. Xu, Z. Yan, C. Dai, and F. Wu, “MEGA-RAG: A retrieval-augmented generation framework with multi-evidence guided answer refinement for mitigating hallucinations of LLMs in public health,” *Frontiers in Public Health*, vol. 13, 2025.
- [17] P. Lewis *et al.*, “Retrieval-augmented generation for knowledge-intensive NLP tasks,” arXivpreprint arXiv:2005.11401, 2021.
- [18]



THANK YOU